



# Database Management Systems

Week 4, Day 4

MongoDB—Aggregate Functions

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# Loading sample Data



The screenshot shows a MongoDB terminal window with a dark blue background. At the top, there are two tabs labeled ">\_ mongosh: MongoDB" and a plus sign icon to the right. Below the tabs, the prompt ">\_MONGOSH" is visible. The command "agree> db.persons.insertMany()" is entered, with the opening curly brace of the array argument highlighted by a blue selection box. On the left side of the terminal, there is a vertical toolbar containing a gear icon, a plus icon, an ellipsis icon, and a funnel icon.

```
>_ mongosh: MongoDB >_ mongosh: MongoDB +
>_MONGOSH
agree> db.persons.insertMany()
```

# Aggregate syntax

```
agree> db.orders.aggregate( [  
  
  // Stage 1: Filter pizza order documents by pizza size  
  {  
    $match: { size: "medium" }  
  },  
  
  // Stage 2: Group remaining documents by pizza name and calculate total quantity  
  {  
    $group: { _id: "$name", totalQuantity: { $sum: "$quantity" } }  
  }  
  
] )
```

```
>_MONGOSH
```

```
> db
```

```
< agree
```

```
> show collections
```

```
< authors
```

```
books
```

```
persons
```

```
users
```

```
agree> db.persons.aggregate([])
```

Aggregate Continue...

# Aggregate Stage

- A stage can filter documents, group documents, and calculate values.
- The documents that are output from a stage are passed to the next stage.

# | Stage

>\_ mongosh: MongoDB



>\_ MONGOSH

agree> db.persons.aggregate([{\$match: {gender: 'male'}}])

>\_MONGOSH

> db.persons.aggregate([{\$match: {name: 'Newman Rodriquez'}}])

```
< {
  _id: ObjectId('675fa2a3be57a37defbdcac2'),
  index: 15,
  name: 'Newman Rodriquez',
  isActive: true,
  registered: 2017-09-19T11:25:18.000Z,
  age: 29,
  gender: 'male',
  eyeColor: 'green',
  favoriteFruit: 'strawberry',
  company: {
    title: 'PHARMACON',
    email: 'newmanrodriquez@pharmacon.com',
    phone: '+1 (906) 591-2086',
    location: {
      country: 'France',
      address: '624 Madeline Court'
    }
  }
}
```



```
>_MONGOSH
> db.orders.aggregate(
  [
    {$match: {size: "medium"}},
    {$group: {_id:"$name", totalquantity: {$sum: "$quantity"}}}
  ])
< {
  _id: 'Pepperoni',
  totalquantity: 20
}
{
  _id: 'Cheese',
  totalquantity: 50
}
{
  _id: 'Vegan',
  totalquantity: 10
}
PizzaHut>
```

# 2 Stages

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# For more details

<https://www.mongodb.com/docs/manual/core/aggregation-pipeline/>

# Stage Operators

**\$match**

**\$group**

**\$project**

**\$sort**

**\$count**

**\$limit**

**\$skip**

**\$out**

# Aggregation Expression

```
>_MONGOSH
> db.persons.aggregate([
  {$group: {_id: "$company.location.country"}}
])
< {
  _id: 'Italy'
}
{
  _id: 'Germany'
}
{
  _id: 'France'
}
{
  _id: 'USA'
}
agree>
```

```
>_MONGOSH
```

```
> db.persons.aggregate([  
  {$group: {_id: "$name", total: {$sum: "$price"}}}  
])
```

## Aggregate Expression

# \$match

- Match specific document using query

`{match: {<query>}}`

## \$match Example

```
> db.persons.aggregate([
  {$match: {age: 40}}
])
< {
  _id: ObjectId('675fa2a3be57a37defbdcaba'),
  index: 7,
  name: 'Anastasia Blake',
  isActive: true,
  registered: 2016-07-01T02:32:46.000Z,
  age: 40,
  gender: 'female',
  eyeColor: 'brown',
```

## \$match continue

```
> db.persons.aggregate([
  {$match: {age: {$lt: 22}}}
])
< {
  _id: ObjectId('675fa2a3be57a37defbdcab3'),
  index: 0,
  name: 'Aurelia Gonzales',
  isActive: false,
  registered: 2015-02-11T04:22:39.000Z,
  age: 20,
```



## \$match continue

```
> db.persons.aggregate([
  {$match: {$and: [{gender: "female"}, {age: {$lt : 22}}]}}
])
< {
  _id: ObjectId('675fa2a3be57a37defbdcab3'),
  index: 0,
  name: 'Aurelia Gonzales',
  isActive: false,
  registered: 2015-02-11T04:22:39.000Z,
  age: 20,
  gender: 'female',
  eyeColor: 'green',
```

## Query Boolean Data

```
agree> db.persons.aggregate(  
  [  
    {$match: {isActive: false}}  
  
  ]  
)
```

# Query Size

```
> db.persons.aggregate(  
  [  
    {$match: {tags: {$size: 2}}}  
  ]  
)  
< {  
  _id: ObjectId('675fa2a3be57a37defbdcab8'),  
  index: 5,  
  name: 'Grace Larson',  
  isActive: true,  
  registered: 2014-04-20T11:37:23.000Z,  
  age: 20,  
  gender: 'female',  
  eyeColor: 'blue',  
  favoriteFruit: 'apple',  
  company: {  
    title: 'OVOL0',  
    email: 'gracelarson@ovolo.com',  
    phone: '+1 (930) 510-3310',  
    location: {  
      country: 'USA',  
      address: '932 Linden Street'  
    }  
  }  
}
```

# Question(s)

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